

Selenium Questions

Easy Level

1. What is Selenium?

Selenium is an open-source tool which is used to automate web browsers. It supports multiple programming languages like Java, JavaScript, python, C++. It comes with different browser support and platform independent.

2. What are selenium components?

We have four components which is in beginning of creation of selenium those are

Selenium IDE, Selenium WebDriver, Selenium RC and Selenium Grid

Due to recent updates of selenium 2 to selenium 3 The selenium RC component was deprecated and now present we have Three components those are

- Selenium IDE
- Selenium WebDriver
- Selenium Grid

- **Selenium Grid:** Distributes and runs tests across multiple machines and browsers
- **Selenium WebDriver:** It is the most popular component of all selenium components, it is used to automate the web browsers
- **Selenium IDE:** is mainly used to record and playback, it is one of the plugins which is available for the browsers where we can download the plugin and record the scripts and rerun the scripts in the same way

3. What are the advantages of using selenium as an automation tool?

It gives some unique benefits over other tool like

- It is a open-source tool
- It comes with multiple programming languages support
- It is platform independent
- It supports multi browser
- We can do parallel testing also

4. What is automation testing?

It is the process of using specialized software to control the execution of tests and compare the results with the expected outcomes.

It helps in reduce the time, cost and effort

5. What are the advantages of Automation testing?

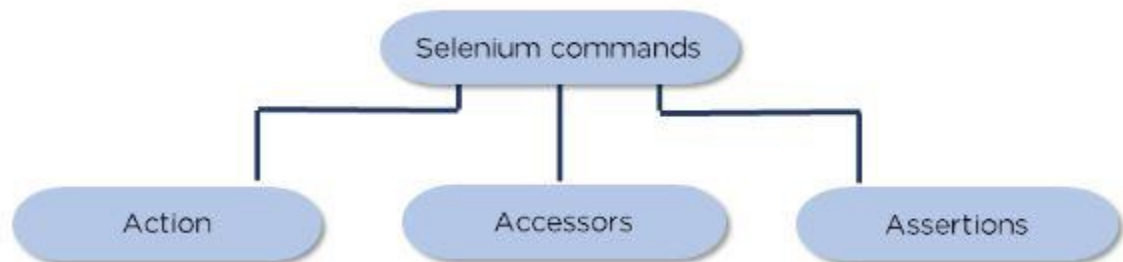
Automation testing has several advantages

- **Saves time and effort:** It speed up the testing by automating the repetitive tasks, like running the same test cases multiple times across different browsers.
- **Improves accuracy:** Automation eliminates human errors, ensuring tests are performing always same way, which is crucial for complex applications.
- **Increases test coverage:** Automating more tests allows you to run more test cases and covers more functionality, helping to ensure your software is thoroughly tested and free of bugs.

6. What is selenese and how it is classified?

Selenese is a language which is used to write test scripts in selenium IDE. It consists of commands to perform actions on web browser, such as clicking on a button, entering text in input field, verifying the presence of elements....

Selenese commands are classified into three categories



- **Actions:** These commands interact with web elements such as click, type, select...
- **Assertions:** This command verifies the state of the application, ensuring it behaves as expected. **Ex:** assertText, assertTitle, assertElementPresent...
- **Accessors:** These commands retrieve the values from the web element and store them in a variable for later use. **Ex:** storeTitle, storeText....

7. What are the limitations of selenium testing?

Selenium testing has some limitations

- **No built-in reports:** we need to use other tools like TestNG or JUnit for test reports and management
- **Web application only:** It needs other tools like Appium for mobile and desktop app testing.
- **Limited image testing:** It does not have strong support for image-based testing
- **Require programming knowledge:** We need to know some programming language to use selenium WebDriver effectively
- **No dedicated tech support:** Being an open-source tool, there isn't a dedicated team to help with the issues.

8. What are the testing types supported by selenium?

Selenium supports several types of testing:

- **Functional testing:** Ensure the application behaves as expected by interacting with the user interface and verify the output.
- **Regression testing:** Verifies the new code changes does not effect the existing functionality of the application
- **Cross browser testing:** Check's the application compatibility across different web browsers. Ex: chrome, safari, Firefox...
- **Smoke testing:** Runs a set of basics tests to ensure the most important functionality of the application works correctly.
- **Integration testing:** Tests how different parts of application works together
- **End-to-End testing:** Simulates real users' scenario to ensure the entire application flow works correctly.

9. What is the same origin policy and how is it handled in selenium WebDriver?

Same-origin-policy:

In the early days of web development, JavaScript could run code from both the same server and other servers. This led to some security problems because harmful scripts from other sites could cause issues.

To fix this, the Same Origin Policy was created. This rule says that JavaScript on a web page can only interact with resources from the same origin (the same domain,

protocol, and port). It prevents scripts from other origins from accessing or changing your website's data.

In simple terms, the Same Origin Policy is a security rule that keeps your web page safe by only allowing JavaScript to interact with the same site it came from.

Selenium handling:

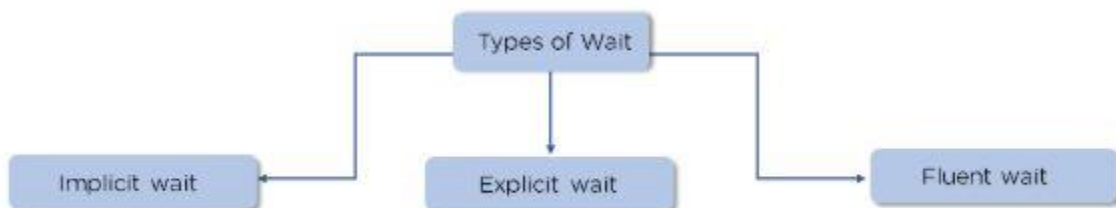
Selenium handles the same-origin-policy by working **within the browser session** where the tests are run. This means:

- **Same origin:** selenium tests interacts with the web application as if it's from the same site, so that policy doesn't block these sites
- **Cross-origin-request:** For testing interactions with different sites (like APIs) selenium relies on the server setup or CORS settings to allow these requests
- **No Bypassing:** Selenium doesn't bypass the same origin policy; it follows the same rules as regular browser interactions.

10. mention the types of Web locators available in selenium?

- By ID
- By name
- By Class Name
- By Tag Name
- By Text
- By Partial Link text
- By CSS selector
- By Xpath

11. What are the types of waits supported by WebDriver



Implicit wait:

If an element has not found immediately, WebDriver will wait up to the specific time before throwing “**noSuchElementException**”.

Ex: driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS)

Explicit wait:

Info: allows us to define a specific wait condition and timeout for individual elements. It is more flexible than implicit wait and can be used to wait for a specific condition to be met.

Use: we create an instance of “**WebDriverWait**” and use “**ExpectedConditions**” to specify the condition to wait for

Ex:

```
WebDriverWait wait = new WebDriverWait(driver, 10)
WebElement element =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("ElementId")))
```

Fluent waits:

Info: provides the ability to wait for a condition with a specific polling and can handle exceptions during waiting. It is more customizable compared to WebDriverWait.

Use: we can set the polling interval and condition to be checked

Ex:

```
Wait<WebDriver> wait = new FluentWait<>(driver)

    .withTimeout(30, TimeUnit.SECONDS)

    .pollingEvery(5, TimeUnit.SECONDS)

    .ignoring(NoSuchElementException.class);

WebElement element = wait.until(driver ->
driver.findElement(By.id("elementId")));
```

12. What is the difference between driver.navigate() and driver.get() in selenium?

In selenium both driver.navigate() and driver.get() functions are used to open web pages, but they differ in their functionality

Driver.get():

It loads a new page in the current browser window

Usage: driver.get("URL")

Behavior: This command waits for the page to fully load before proceeding to the next command. It is Stright way to open a web page

Driver.navigate().to():

Purpose: Also load a new web page but as a part of API

Usage: driver.navigate().to("URL")

Behavior: similar to driver.get() but It is part of "navigate "interface, which provides additional capabilities like back, forward and refresh...This command can see as more flexible when seen with other commands.

13. Mention the types of navigation commands?

- **Navigate().to()**

Loads a new web page in the current browser window

Ex: driver.navigate().to("www.url.com")

- **Navigate().back()**

Stimulates the browser back button, navigating to the previous page in the browser history.

Ex: driver.navigate().back();

- **Navigate().forward()**

Stimulates the browser forward button, navigate to the next page in the browser's history (if available)

Driver.navigate().forward()

- **Navigate().refresh()**

Refresh the current web page

Ex: driver.navigate().refresh()

14. What is the difference between driver.close() and driver.quit()?

Driver.close()

This command closes the browser current window, if multiple windows are opened the current window will be closed only.

Driver.quit()

When quit() is called on the driver instance and there are one or more browser instances are opened, it closes all the open browser windows

15. What makes Selenium such a widely used testing tool? Give reasons?

We can tell all the selenium advantages

16. Why is it advised to select Selenium as a testing tool for web applications or systems?

- Selenium is an open-source, portable tool and freeware.
- It supports many operating systems like Linux, Unix, Macintosh and Windows.
- Selenium supports Chrome, opera, safari, and internet explorer.
- It supports many languages like Perl, Java, Python, Ruby, Groovy, JavaScript, VB Script, etc.
- Selenium can be used for iPhone, Android, and Blackberry based on application testing.
- It helps integrate it with ANT or Maven framework for source code compilation.
- It requires less CPU and RAM consumption for script execution.

17. What is an exception test case in selenium?

It is a test case that we are expecting an exception inside a test case.

If we write a test case that, it should throw an exception, then we can use the test case in test automation by specifying exception parameters.

```
import org.testng.annotations.Test;

import org.openqa.selenium.NoSuchElementException;

public class ExceptionTest {

    @Test(expectedExceptions = NoSuchElementException.class)

    public void testForException() {

        WebElement element =
driver.findElement(By.id("nonExistentElement"));

    }

}
```

18. How to wait until a web page has been loaded completely in Selenium?

One approach is using the “implicit wait” command in selenium, which instructs the WebDriver to wait a certain amount of time before throwing an error, if the element is not found or loaded.

Another option is used to “Explicit wait” command to wait for a specific element to appear on the page before proceeding with the script.

19. What is selenium WebDriver?

Selenium WebDriver is a popular tool for automating web browsers and has program interface

20. Is Selenium WebDriver is a library?

No, Selenium WebDriver is not a library. It is a **web automation Framework**

Explanation: While selenium WebDriver includes libraries for different programming languages, calling it just a library is an understatement. It is a comprehensive web automation framework that provides the necessary tools and infrastructure for automating web browser interactions, supporting a wide range of browsers and programming languages, and facilitating robust and efficient web application testing.

21. Which browsers/drivers are supported by selenium WebDriver?

It supports multiple web browsers including

- Google Chrome
- Mozilla Firefox
- MicroSoft Edge
- Safari
- Opera

22. What will happen if I execute this command? `driver.get`

This command will instruct the Selenium WebDriver to open the specified URL in the web browser.

23. What is an alternative option to `driver.get()` method to open an URL in Selenium Web Driver?

An alternative option to the `driver.get()` method to open an URL in Selenium Web Driver is to use the `driver.navigate().to()` method.

24. Is it possible to test APIs or web services using Selenium Webdriver?

No, it is not possible. Because Selenium WebDriver is designed for automating web browsers, but we can achieve this by integrating third party libraries like restAssured.

25. How can we move to the nth-child element using XPath?

We can move to the nth-child element using XPath by using the following expression: `(//parent-element/*)[n]`.

26. How can we type text in a text box using selenium?

We can use `sendKeys()` method to type text in input fields

27. How to click on a hyperlink in Selenium?

```
driver.findElement(By.linkText("Today's deals")).click();
```

Intermediate Level Selenium Questions

1. How to scroll down a page using JavaScript?

ScrollBy() method is used to scroll down the web page

General syntax:

```
public class ScrollExample {  
  
    public static void main(String[] args) {  
  
        driver.get("https://www.example.com");  
  
        // Create JavascriptExecutor instance  
  
        JavascriptExecutor js = (JavascriptExecutor) driver;  
  
        // Scroll down by a specific amount (e.g., 500 pixels)  
  
        js.executeScript("window.scrollBy(0, 500);");  
  
        // Or scroll to the bottom of the page  
  
        js.executeScript("window.scrollTo(0, document.body.scrollHeight);");  
  
        // Or scroll to a specific element  
  
        WebElement element = driver.findElement(By.id("elementId"));  
  
        js.executeScript("arguments[0].scrollIntoView(true);", element);  
  
    }  
}
```

2. How to scroll using selenium?

By using “**action**” class in selenium

Ex:

```
Actions actions = new Actions(driver);  
actions.moveToElement(scrollableElement)
```

3. How to assert the title of the page?

Get the title of the web page by using driver.getTitle()

And store in a variable and do the comparing with the actual title text

4. How to mouse hover on a web element?

Action class is used to do the hover actions in selenium WebDriver

```
Actions action = new Actions(driver);  
Actions.moveToElement(WebElement).perform();
```

5. How to retrieve CSS properties of an element?

GetCssValue() method is used to get the CSS property of any web element

General Syntax:

```
Driver.findElementBy(locatorOfWebElement).getCssValue("nameOfCssAttribute")
```

6. What is POM?

Page Object Mode is a design pattern in test automation that helps make test code more organized and easier to maintain.

It works by creating a separate class file for each web page. This class contains

- **Locators**
- **Methods**

By using POM, we keep the code interacting with a page separate from our test logic. This makes our tests cleaner and easier to maintain, because we don't have to rewrite the code for page interactions every time we write a new test. Instead, we use the methods in the page object class.

For example, if we have a login page, we will create a "**login page**" class that has methods for entering the username, password and clicking on login button. Our test script uses this class to perform actions on the page, which makes our test code more straightforward and reusable

7. What is object Repository

An object repository is a centralized location where testers can store all the web elements, like buttons, text boxes, links used in test automation framework.

8. Can Captcha be automated?

No, Selenium cannot automate Captcha. Well, the whole concept of Captcha is to ensure that bots and automated programs don't access sensitive information - which is why Selenium cannot automate it. The automation test engineer must manually type the captcha while other fields can be filled automatically

9. What is exactly meant by a WebElement in Selenium, and how is it used?

The WebElement is an interface in Selenium which is used to represent an HTML element on a web page. It provides methods to interact with the web elements, such as clicking, entering text, and getting the value. It is used to identify and manipulate the web elements in the automation testing process.

ADVANCED LEVEL QUESTIONS

1. Is there a way to type in a textbox without using sendKeys()?

Yes, there are alternative methods to type text in the text box without using sendkeys

- **JavaScript Executor**

```
WebElement inputField = driver.findElement(By.id("textboxId"));
JavascriptExecutor js = (JavascriptExecutor) driver;
js.executeScript("arguments[0].value = 'Some text';", inputField);
```

- **Actions Class**

```
WebElement inputField = driver.findElement(By.id("textboxId"));
Actions actions = new Actions(driver);
actions.moveToElement(inputField).click().sendKeys("Some text").perform();
```

2.

10. How does Selenium handle Windows-based pop-ups?

Selenium was designed to handle web applications. Windows-based features are not natively supported by Selenium. However, third-party tools like AutoIT, Robot, etc can be integrated with Selenium to handle pop-ups and other Windows-based features.

11. How to take screenshots using WebDriver?

TakeScreenshot interface used to capture the screenshots in WebDriver
GetScreenshotAs() method can be used to save the screenshot

General code:

```
File scrFile = ((TakeScreenshot)driver).getScreenshotAs(outputType.FILE);
```

12. What is the difference between getWindowHandles() and getWindowHandle()?

getWindowHandle() method returns the unique identifier of the current browser window, while getWindowHandles() method returns a set of unique identifiers of all the browser windows opened by WebDriver.

13.